

UNO

Input file: **standard input**
Output file: **standard output**
Time limit: 1 second
Memory limit: 1024 megabytes

Troubadour_Ggmz and Fengmi123 are playing a special UNO game. UNO cards have four colors: red, yellow, green, and blue, denoted by **R**, **Y**, **G**, and **B**, respectively. Each color contains n cards, with ranks being any integers from 1 to n ; there may be two cards with the same color and rank. The whole deck has $4n$ cards.

At the beginning of the game, Troubadour_Ggmz arranges all $4n$ cards in a row, then Fengmi123 can perform the following operation any number of times: choose two adjacent cards, and if they are the same color or the same rank, swap their positions.

The goal of the game is to rearrange all cards into a state that is both color-ordered and rank-ordered, i.e.:

- The color order is $R < Y < G < B$, meaning all red cards appear before all yellow cards, all yellow cards appear before all green cards, and all green cards appear before all blue cards;
- For cards of the same color, the sequence of ranks must be non-decreasing.

Fengmi123 wants to know whether she can achieve the target state through a sequence of allowed operations (possibly zero).

Input

There are multiple test cases. The first line contains an integer t ($1 \leq t \leq 1000$), the number of test cases. For each test case:

The first line contains a positive integer n ($1 \leq n \leq 1000$).

Then $4n$ lines follow, each containing a positive integer x_i and a character c_i , where $1 \leq x_i \leq n$, $c_i \in \{R, Y, G, B\}$, indicating the rank and color of the i -th card in order. It is guaranteed that each color appears exactly n times, but cards with the same color and rank may appear multiple times.

It is guaranteed that the sum of n over all test cases does not exceed 1000.

Output

For each test case, if the target state can be reached, output a line containing “YES”, otherwise output a line containing “NO”.

You may output the answer in any case (upper or lower). For example, the strings “yEs”, “yes”, “Yes”, and “YES” will all be recognized as positive responses.

Example

standard input	standard output
2	YES
2	NO
1 R	
2 Y	
2 R	
1 G	
1 Y	
2 B	
2 G	
1 B	
2	
1 R	
2 Y	
1 Y	
2 B	
2 R	
1 G	
2 G	
1 B	